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59.

S. No	Liquid	Boiling points (^{0}C)	ΔH (kJ mol^{-1})
1.	Ethanol	78.4	+ 42.4
2.	Toluene	110.6	+ 35.2

SOLUTION :

For ethanol :

Given :

$T_b = 78.4^0$ C = (78.4 + 273)

= 351.4 K

ΔH_V(ethanol) = + 42.4 kJ mol^{-1}

$$\Delta S_V = \frac{\Delta H_V}{T_b}$$

$$\Delta S_V = \frac{+ 42.4 \text{ kJ mol}^{-1}}{351.4 \text{ K}}$$

$$\Delta S_V = \frac{+ 42400 \text{ J mol}^{-1}}{351.4 \text{ K}}$$

ΔS_V = +120.66 J K^{-1} mol^{-1}

For Toluene :

Given :

$T_b = 110.6^0$ C = (110.6+273)

= 383.6 K

ΔH_V (toluene) = + 35.2 kJ mol^{-1}

$$\Delta S_V = \frac{\Delta H_V}{T_b}$$

$$\Delta S_V = \frac{+ 35.2 \text{ kJ mol}^{-1}}{383.6 \text{ K}}$$

$$\Delta S_V = \frac{+ 35200 \text{ J mol}^{-1}}{383.6 \text{ K}}$$

ΔS_V = + 91.76 J K^{-1} mol^{-1}

60.

Solution :

Given :

ΔH = 30.56 kJ mol^{-1}

= 30560 J mol^{-1}

$\Delta S = 6.66 \times 10^{-3}$ kJ$K^{-1}mol^{-1}$

T = ? at which ΔG=0

$\Delta G = \Delta H - T\Delta S$

$0 = \Delta H - T\Delta S$

$$T = \frac{\Delta H}{\Delta S}$$

$$T = \frac{30.56 \text{ kJ mol}^{-1}}{6.66 \times 10^{-3} \text{ kJK}^{-1} \text{ mol}^{-1}}$$

T = 4589 K

(i) At 4589K ; ΔG = 0 the reaction is in equilibrium.

(ii) at temperature below 4598 K , $\Delta H > T \Delta S$

$\Delta G = \Delta H - T \Delta S > 0$, the reaction in the forward direction, is non spontaneous. In other words the reaction occurs in the backward direction.